

Application Program Development

Ch 01: Introduction to Computers and Programming

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- Python must be installed and configured prior to use (see Getting Started)
- Python interpreter can be used in two modes:
 - **Interactive mode**: enter statements on keyboard
 - **Script mode**: save statements in Python script (file) and pass to interpreter

Concept

The Python interpreter can run Python programs that are saved in files or interactively execute Python statements that are typed at the keyboard. Python comes with a program named IDLE that simplifies the process of writing, executing and testing programs.



Interactive Mode

- When you start Python in interactive mode, you will see a prompt
 - Indicates the interpreter is waiting for a Python statement to be typed
 - Prompt reappears after previous statement is executed
 - Error message displayed if you incorrectly type a statement
- Good way to learn new parts of Python
 - Or test out ideas when writing / designing applications

```
C:\Users\etamustudent> python3
```

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit
```

```
Type "help", "copyright", "credits" or "license" for more information.
```

```
>>> name = input('What is your name? ')
```

```
What is your name? Schrodinger
```

```
>>> print('Hello', name)
```

```
Hello Schrodinger
```

```
>>>
```



Script Mode

- Statements entered in interactive mode are not saved as a program
- To have a program use script mode
 - Save a set of Python statements in a file
 - The filename should have the .py extension
 - To run the file, or script, type the following at the OS command line

```
C:\Users\etamustudent> python3 q02.py  
What is your name? Schrodinger  
Hello Schrodinger!
```



The IDLE Programming Environment

- **IDLE** (Integrated Development Program): single program that provides tool to write, execute and test a program (Appendix B)
 - Automatically installed when Python language is installed
 - Runs in interactive mode
 - Has built-in text editor with features designed to help write Python programs
 - A basic IDE, but recommend using VSCode, PyCharm



References



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